

GECE10069 (561085) F25: Introduction to Programming (C++)

Lab 5 : Literals & Variables



What you will learn from Lab 5

In this laboratory, you will further explore special output symbols and practice using variables in your programs.

TASK 4-1 : #DEFINE V.S. CONST NUMBER

- ✓ Figure out why coercion happened
- ✓ Try two different PI_1 & PI_2

```
#include <iostream>
#include <cmath>
#include <cstdlib>
#include <ctime>
using namespace std;

#define PI 3.1415926

int main() {
    int a = 5, b = 2;
    cout << "a / b = " << a / b << endl;

    double x = 5.5;
    // cout << "x % 2 = " << x % 2 << endl;

    int angle = 90;
    cout << "sin(90) = " << sin(angle) << endl;

    // 沒有 srand() , rand() 每次結果一樣
    cout << "Random number (1~100): " << rand() % 100 << endl;

    double y = 2.5;
    cout << "floor(2.5) = " << floor(y) << endl;
    cout << "ceil(2.5) = " << ceil(y) << endl;
    cout << "round(2.5) = " << round(y) << endl;

    return 0;
}
```

EXERCISE 5-1: SORT THREE NUMBERS

Write a C++ program that reads three integers from the user, and then outputs them in ascending order.

Requirements

- Your program should always print the three numbers in non-decreasing order.

Sample Input - 1
7 4 10
Sample Output - 1
4 7 10
Sample Input - 2
77 77 28
Sample Output - 2
28 77 77
Sample Input - 3
100 0 -100
Sample Output - 3
-100 0 100

EXERCISE 5-2: TAXI FARE CALCULATOR

Write a C++ program to calculate a taxi fare based on the distance.

Requirements

Input: one floating-point number (travel distance, unit = 0.1 km).

Rules:

1. **Base fare:** \$70 if distance ≤ 1.5 km.

2. If distance > 1.5 km:

For every additional 0.1 km, charge \$5.

(Example: distance = 2.0 km $\rightarrow$ extra = 0.5 km $\rightarrow$ Total = \$70 + \$25 = \$95.)

3. If distance > 20.0 km:

After calculating the total, add a 10% surcharge (no rounding).

Output: the final fare (can be decimal).

Sample Input - 1
1.5
Sample Output - 1
70
Sample Input - 2
2.0
Sample Output - 2
95
Sample Input - 3
21.0
Sample Output - 3
1149.5